An Ablation Study of Auxiliary Inputs for Lightweight Neural Upscaling in Real-Time Rendering

Zhanbolot Kydyraliev and Amin Soltobekov

Affiliation to be confirmed

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Abstract

This manuscript is in preparation. The abstract will be written after the research question, experimental protocol, and results are finalized.

1 Introduction

Neural upscaling reconstructs a higher-resolution image from lower-resolution rendering data. A real-time renderer can also provide auxiliary signals, such as depth and motion vectors, that may help reconstruction. Each additional input has a potential quality benefit and a cost in data preparation, memory, and runtime. This study is planned as a controlled ablation of selected auxiliary inputs for a lightweight upscaler. The exact model, inputs, hardware, and evaluation conditions will be fixed in the experiment protocol before results are collected.

We address five research questions about the individual and combined contributions of depth and surface normals, the effect of motion-aligned history, the measured costs of these variants, their performance on unseen scenes, and where in an image the inputs help or hurt. The questions and study design are given in Section 3.1.

2 Background and Related Work

The following work motivates the use of temporal history, rendering buffers, and sample reconstruction in the proposed study.

2.1 Temporal Anti-Aliasing and Supersampling

Low-resolution rendering can produce visible aliasing because rendered pixels sample the scene at discrete positions [1]. Karis's temporal anti-aliasing method for Unreal Engine 4 combines samples from successive frames after shifting the sampling positions between frames [2]. Motion vectors reproject the previous result into the current frame. Inaccurate motion vectors can smear the result, while stale history can cause ghosting; Karis uses neighborhood clipping to limit the influence of incompatible history. These observations motivate our comparison between unaligned and motion-aligned history. That comparison tests the effect of alignment within our proposed upscaler, not the full temporal anti-aliasing method.

2.2 Neural Supersampling for Real-Time Rendering

Xiao et al. [1] present neural supersampling for real-time rendering. Their method reconstructs a high-resolution frame from low-resolution rendered color and uses depth, motion vectors, and information from previous frames. Motion vectors align earlier frames with the current frame, and the method learns to use the aligned history where it is useful. Their work demonstrates a system that combines current-frame rendering buffers with temporal information. Our proposed study asks a narrower question: how quality and measured cost change across selected input variants under a shared experimental protocol. In particular, V4 and V5 compare history without and with motion alignment; they do not measure the effect of motion vectors independently of temporal history.

2.3 Pixels as Samples

Smith [3] explains that a pixel represents a sample of an underlying image rather than a small square of uniform color. Upscaling can therefore be viewed as reconstruction from samples. This perspective makes classical interpolation, such as bicubic interpolation, a useful candidate baseline for a learned method. Separately, the proposed study will render low-resolution inputs

natively at their intended input resolution, rather than create them by filtered downsampling of high-resolution frames. This choice is intended to retain the aliasing produced by the selected rendering process; it is a decision about our experimental data, not a claim made by Smith.

3 Method

3.1 Research Questions and Study Design

We study how auxiliary rendering inputs affect a lightweight neural upscaler. We ask five research questions.

RQ1 (individual contribution) Relative to color-only input, how do depth and surface normals each affect reconstruction quality, and how does motion-aligned history compare with unaligned history?

RQ2 (combination) Do the quality gains from depth and normals overlap in the single-frame setting, and what is the incremental effect of adding them together to motion-aligned history?

RQ3 (cost) How do the selected runtime and memory measurements vary across the variants on the provisional target GPU, relative to their reconstruction quality?

RQ4 (generalization) Do quality differences between variants persist on held-out scenes, and how do they vary between those scenes?

RQ5 (where the inputs help) Where in the image do auxiliary inputs help or hurt, for example at geometric edges, on thin structures, and in regions newly revealed by motion?

Hypotheses. Before running experiments, we state expectations that the study can confirm or refute.

H1 Adding depth improves quality over color-only input.

H2 Adding surface normals improves quality over color-only input.

H3 Depth and normals are partly redundant, so using both improves quality by less than the sum of their individual gains.

H4 History warped with motion vectors improves quality over history used without alignment. Incorrect motion can smear temporal results [2], while motion-based alignment is used in neural supersampling [1].

H5 Improvements of selected auxiliary-input variants over their matched baselines persist on scene-disjoint test scenes.

Study design. We plan to use a shared base architecture while varying the inputs. A shared design does not imply identical parameter counts: changing the number of input channels changes the first layer, and temporal variants may require additional processing. The final architecture and treatment of temporal history will be fixed before experimentation. Training data, loss, optimizer, and schedule should be held constant across variants where applicable. We plan to use multiple random seeds; their number remains to be set. Parameter counts and computational complexity will be reported so that quality differences are not attributed solely to the inputs if model capacity also differs. Possible controls include matching parameter budgets or adding a capacity-matched comparison, subject to protocol finalization.

Table 1 lists the core variants. V1 versus V0 isolates depth, and V2 versus V0 isolates normals. V3 versus V1 estimates the incremental effect of normals with depth, while V3 versus V2 estimates the incremental effect of depth with normals; together with V0, these comparisons test overlap in the single-frame setting. V4 versus V0 estimates the effect of adding unaligned history, while V5 versus V4 tests motion alignment of that history, provided the temporal model and history inputs are otherwise held constant. V6 versus V5 estimates the combined addition of depth and normals to aligned history; V6 versus V3 compares the combined temporal variant with its single-frame counterpart. These comparisons do not independently identify every depth, normal, and temporal interaction. In particular, the matrix does not isolate motion vectors without history or the individual effects of depth and normals when aligned history is present. Variants that add only depth or only normals to aligned history could be considered as optional extensions if the study scope permits.

Training, validation, and test scenes will be disjoint at the scene level; frames from a test scene will not be used for model fitting or protocol selection. Validation scenes will support model and protocol choices, while held-out test scenes will support the reported comparisons. Scenes with a defined distribution shift would require a separate out-of-distribution test; being unseen during training alone does not establish such a shift. RQ1 and RQ2 use the variant comparisons above; RQ3 compares quality with the selected cost measurements; RQ4 compares variants across held-out scenes; and RQ5 examines spatial error patterns, including error maps. Candidate metrics and cost dimensions are discussed in Section 3.3. The temporal variants are a planned extension, and their implementation and the final protocol remain to be fixed.

Table 1: Planned core input variants (V0 is the color-only baseline).

ID	Group	Color	Depth	Normals	History
V0	Single frame	yes	–	–	–
V1	Single frame	yes	yes	–	–
V2	Single frame	yes	–	yes	–
V3	Single frame	yes	yes	yes	–
V4	Temporal	yes	–	–	unaligned
V5	Temporal	yes	–	–	motion aligned
V6	Temporal	yes	yes	yes	motion aligned

3.2 Model, Inputs, and Data

The architecture, input preprocessing, scenes, resolution, and upscaling factor are to be selected. The training, validation, and test scene lists and the method for producing aligned or unaligned history will be fixed before any experiments are run.

3.3 Evaluation Protocol

Candidate image-quality measures include PSNR and SSIM; a perceptual metric may be added if it suits the final reference images. These measures would support RQ1, RQ2, and RQ4. A temporal consistency measure may be useful for RQ5 and the temporal comparisons, provided it can distinguish instability from genuine scene motion. Error maps and selected image regions can also support RQ5. The final measures, aggregation across frames and scenes, and statistical treatment across seeds will be specified before testing.

For RQ3, possible cost measures include neural inference latency and GPU memory consumption. Auxiliary-buffer preparation, history warping and storage, data transfers, and total upscaling-pipeline time may also matter for deployment. The protocol must specify which of these are measured, how their boundaries are defined, and the target hardware and timing procedure. Any costs outside those boundaries will be identified as unmeasured, so a quality–cost

comparison is not mistaken for a complete pipeline benchmark. The measurement protocol and any interpolation baseline remain to be fixed before the experiments.

4 Results

Results have not yet been collected.

4.1 Contribution of Individual and Combined Inputs (RQ1, RQ2)

4.2 Runtime and Memory Cost (RQ3)

4.3 Generalization to Unseen Scenes (RQ4)

4.4 Where the Inputs Help or Hurt (RQ5)

5 Discussion

This section will answer RQ1 to RQ5 in turn, state whether hypotheses H1 to H5 were supported by the results, and discuss the limitations of the study.

6 Conclusion

Conclusions will be written after the analysis.

References

[1] Lei Xiao et al. "Neural Supersampling for Real-time Rendering". In: *ACM Transactions on Graphics* 39.4, 142 (2020). DOI: 10.1145/3386569.3392376.

[2] Brian Karis. *High Quality Temporal Supersampling*. Presentation at SIGGRAPH 2014 (Advances in Real-Time Rendering course), Epic Games. Slides. 2014. URL: https://de45xmedrsdbp.cloudfront.net/Resources/files/TemporalAA_small-59732822.pdf.

[3] Alvy Ray Smith. *A Pixel Is Not A Little Square, A Pixel Is Not A Little Square, A Pixel Is Not A Little Square! (And a Voxel is Not a Little Cube)*. Technical Memo 6. Microsoft, 1995. URL: https://alvyray.com/Memos/CG/Microsoft/6_pixel.pdf.